c11M6

SEQUENCES AND SERIES

Sequence : numbers written in a definite order, following some rule.

Ex: $2, 4, 8, 16, \ldots$; $a_1, a_2, a_3, \ldots, a_n$ $\Rightarrow$ general term is T_n (or a_n).

$\rightarrow$ **Finite** sequence : last term hota hai. $\rightarrow$ **Infinite** : kabhi khatam nahi hota.

Series : terms of a sequence joined by "+" sign.

$T_1 + T_2 + T_3 + \cdots + T_n$; sum of first n terms is written as $\boxed{S_n}$

Progression : sequence whose T_n follows a pattern / formula.

$\rightarrow$ Ex: A.P., G.P., H.P. $\rightarrow$ Every progression is a sequence but not every sequence is a progression (prime numbers $2, 3, 5, 7, \ldots$ ka koi formula nahi).

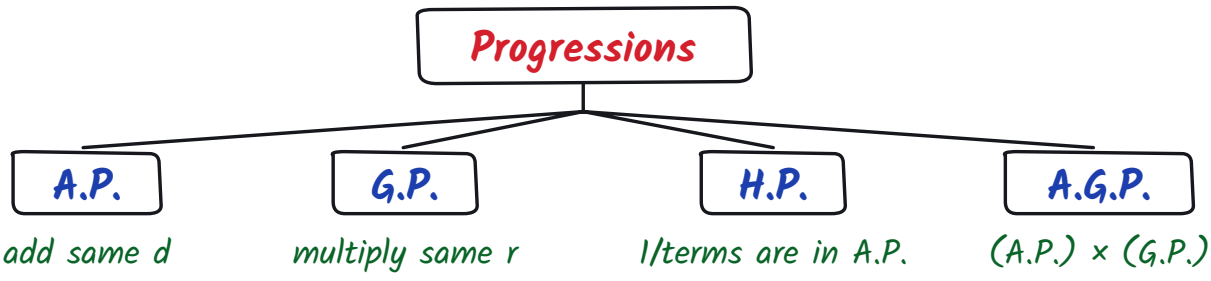
Defining Series and Progression

$\boxed{\text{☆ } T_n = S_n - S_{n-1}\ (n \geq 2) \ ; \quad T_1 = S_1}$ S_n diya ho toh T_n aise nikalo.

Ex: $S_n = n^2 + 2n \Rightarrow T_1 = 3, \quad T_n = n^2 + 2n - (n-1)^2 - 2(n-1) = 2n + 1$

$T_1 = S_1$ pehle alag se check karo. $n = 1$ pe formula $2(1) + 1 = 3$ match hua, so $T_n = 2n + 1$ for all n. Agar match na kare toh T_1 alag likhna.

Types of Progression :



Pattern pehchano (kaunsa type hai?)

$\rightarrow$ $T_n - T_{n-1}$ constant $\Rightarrow$ A.P. ; T_n / T_{n-1} constant $\Rightarrow$ G.P.

$\rightarrow$ $1/T_n$ ka difference constant $\Rightarrow$ H.P. ; (AP)$\times$(GP) $\Rightarrow$ A.G.P.

$\rightarrow$ differences khud A.P./G.P. bante hain $\Rightarrow$ Method of Difference

Arithmetic Progression (A.P.) : $T_n - T_{n-1} = d$ (constant)

a = first term ; d = common difference ; l = last term ; n = number of terms

☆☆ $T_n = a + (n-1)d$; T_n from end $= l - (n-1)d$

☆ $S_n = \frac{n}{2}[2a + (n-1)d] = \frac{n}{2}(a + l)$

n odd ho toh middle term $T_{\frac{n+1}{2}}$ hota hai aur $S_n = n\times$ (middle term). Also $d = S_n - 2S_{n-1} + S_{n-2}$.

Selection of terms (jab sum diya ho)

→ 3 terms : $a-d,\ a,\ a+d$ → 5 terms : $a-2d,\ a-d,\ a,\ a+d,\ a+2d$

→ 4 terms : $a-3d,\ a-d,\ a+d,\ a+3d$ (yahan common diff = $2d$)

Important properties

(i) Har term me same number add / subtract / multiply / divide ($\neq 0$) karo $\Rightarrow$ phir bhi A.P.

(ii) T_n linear in n $\iff$ A.P. ; $S_n = An^2 + Bn$ (no constant) $\iff$ A.P. with $d = 2A,\ a = A + B$

(iii) Equidistant terms: $T_k + T_{n-k+1} = T_1 + T_n$; a, b, c in A.P. $\iff 2b = a + c$

(iv) $T_m = n,\ T_n = m \Rightarrow T_{m+n} = 0$; $S_m = S_n \Rightarrow S_{m+n} = 0$

(v) $T_m = \frac{1}{n},\ T_n = \frac{1}{m} \Rightarrow S_{mn} = \frac{mn+1}{2}$

☆☆ (vi) $\frac{S_n}{S'_n} = \frac{An+B}{Cn+D} \Rightarrow \frac{T_n}{T'_n} = \frac{A(2n-1)+B}{C(2n-1)+D}$ n ki jagah (2n-1)

Ex: $\frac{S_n}{S'_n} = \frac{7n+1}{4n+27}$, find $\frac{T_{11}}{T'_{11}}$.

$\Rightarrow n = 2(11) - 1 = 21 \Rightarrow \frac{7(21)+1}{4(21)+27} = \frac{148}{111} = \boxed{\frac{4}{3}}$

Ex: Sum of all 2-digit numbers divisible by 7 : $14, 21, \ldots, 98$

$n = \frac{98-14}{7} + 1 = 13 \Rightarrow S = \frac{13}{2}(14+98) = \boxed{728}$

Arithmetic Mean (A.M.)

$A = \dfrac{a+b}{2}$; n A.M.s insert karo a aur b ke beech :

$\boxed{d = \dfrac{b-a}{n+1}}$; sum of the n A.M.s $= n \cdot \dfrac{a+b}{2}$ ($n \times$ single A.M.)

Ex: Teen numbers A.P. me, sum $= 33$, product $= 1155$. $a-d, a, a+d$ lo :

$3a = 33 \Rightarrow a = 11$; $11(121 - d^2) = 1155 \Rightarrow d^2 = 16$ $\Rightarrow$ numbers $7,\ 11,\ 15$

Ex: $S_n = 3n^2 + 5n$ aur $T_m = 164$. Find m. $\Rightarrow$ $T_n = S_n - S_{n-1} = 6n + 2$

$6m + 2 = 164 \Rightarrow m = 27$ ($d = 2A = 6$ ✓)

More A.P. tricks (Main + Advanced)

☆☆ (i) $\boxed{S_p = q,\ S_q = p \Rightarrow S_{p+q} = -(p+q)}$

(ii) $\dfrac{S_n}{S_m} = \dfrac{n^2}{m^2} \Rightarrow \dfrac{T_n}{T_m} = \dfrac{2n-1}{2m-1}$ (yahan $a = \frac{d}{2}$)

(iii) $2n+1$ terms ho toh : $\boxed{S_{\text{odd places}} : S_{\text{even places}} = (n+1) : n}$

(iv) S_n diya ho toh : $a = S_1$; $d = S_2 - 2S_1$

Trap : S_n = given, par n ke do answer

Ex: $24, 20, 16, \ldots$ ke kitne terms ka sum 72 hai ? $\Rightarrow$ $\dfrac{n}{2}[48 - 4(n-1)] = 72$

$n^2 - 13n + 36 = 0 \Rightarrow n = 4$ ya 9 ($0, -4, -8$ ka sum -12)

d negative ho toh dono roots valid ho sakte hain!

Sum of absolute values $\sum |T_r|$

Sign change dhundo : $T_k > 0,\ T_{k+1} < 0 \Rightarrow \boxed{\sum_{r=1}^{n} |T_r| = 2S_k - S_n}$

Ex: $20, 17, 14, \ldots$ ke pehle 12 terms ke $|T_r|$ ka sum : $T_r = 23 - 3r$

$T_7 = 2 > 0,\ T_8 = -1 < 0 \Rightarrow S_7 = 77,\ S_{12} = 42 \Rightarrow 2(77) - 42 = 112$

Grouped series $1;\ 2, 3;\ 4, 5, 6;\ 7, 8, 9, 10; \ldots$

☆ n-th group me n terms ; pehla term $\boxed{\dfrac{n(n-1)}{2} + 1}$

Sum of n-th group $= \boxed{\dfrac{n(n^2+1)}{2}}$ (check : $n = 3$: $4 + 5 + 6 = 15 = \frac{3 \cdot 10}{2}$)

Trick : pehle k groups me $\frac{k(k+1)}{2}$ terms hote hain.

Geometric Progression (G.P.) : $\frac{T_n}{T_{n-1}} = r$ (constant, $r \neq 0$)

a = first term ; r = common ratio

☆☆ $T_n = ar^{n-1}$; T_n from end $= l\left(\frac{1}{r}\right)^{n-1}$

☆ $S_n = a\frac{r^n - 1}{r-1}\ (r > 1)$; $a\frac{1-r^n}{1-r}\ (r < 1)$; $na\ (r = 1)$

☆ $S_\infty = \frac{a}{1-r}$ only when $|r| < 1$

Selection of terms

→ 3 terms : $\frac{a}{r}, a, ar$ → 5 terms : $\frac{a}{r^2}, \frac{a}{r}, a, ar, ar^2$

→ 4 terms : $\frac{a}{r^3}, \frac{a}{r}, ar, ar^3$ (common ratio r^2)

Important properties

(i) Har term ko same non-zero number se multiply / divide karo ⇒ phir bhi G.P.

(ii) Reciprocals $\frac{1}{T_n}$ bhi G.P. (ratio $\frac{1}{r}$) ; T_n^k bhi G.P. (ratio r^k)

(iii) a, b, c in G.P. $\iff b^2 = ac$; $T_k T_{n-k+1} = T_1 T_n$ (equidistant ka product)

(iv) Product of first n terms : $P_n = a^n r^{\frac{n(n-1)}{2}}$

(v) Har k-th term lo ⇒ G.P. with ratio r^k ; $a_i > 0$ in G.P. $\iff \log a_i$ in A.P.

☆ (vi) Agar numbers A.P. aur G.P. dono me ho ⇒ $a = b = c$ (constant sequence)

Geometric Mean (G.M.)

$G = \sqrt{ab}$; n G.M.s insert karo a aur b ke beech :

$r = \left(\frac{b}{a}\right)^{\frac{1}{n+1}}$; product of the n G.M.s $= (\sqrt{ab})^n$

Ex: G.P. me $T_2 = 6$, $T_5 = 48$. T_n aur S_6 ? ⇒ $r^3 = \frac{48}{6} = 8 \Rightarrow r = 2,\ a = 3$

$T_n = 3 \cdot 2^{n-1}$; $S_6 = 3 \cdot \frac{2^6 - 1}{2 - 1} = 189$

Ex: Recurring decimal $0.\overline{36} = 0.36 + 0.0036 + \cdots = \dfrac{0.36}{1 - 0.01} = \dfrac{4}{11}$

Ex: $0.7 + 0.77 + 0.777 + \ldots$ (n terms) $= \dfrac{7}{9}\left[n - \dfrac{1 - 10^{-n}}{9}\right]$

($\frac{7}{9}(0.9 + 0.99 + \ldots)$ likho, $0.9 = 1 - 0.1$, $0.99 = 1 - 0.01$, ... phir G.P. lagao.)

☆ **Ex:** $x = 1 + a + a^2 + \ldots$, $y = 1 + b + b^2 + \ldots$ ($|a|, |b| < 1$) toh $1 + ab + a^2b^2 + \cdots = \dfrac{xy}{x + y - 1}$

Reason : $a = \frac{x-1}{x}$, $b = \frac{y-1}{y}$ nikalo, phir $\dfrac{1}{1 - ab}$ me daalo.

Ex: $6 + 66 + 666 + \ldots$ (n terms) $= \dfrac{6}{9}(9 + 99 + 999 + \ldots)$

$= \dfrac{2}{3}\left[\dfrac{10(10^n - 1)}{9} - n\right] = \dfrac{2\,(10^{n+1} - 10 - 9n)}{27}$

More G.P. tricks (Main + Advanced)

(i) $\boxed{S_{2n} = S_n(1 + r^n)}$; $S_{3n} = S_n(1 + r^n + r^{2n})$ Ex: $\frac{S_{2n}}{S_n} = 9 \Rightarrow r^n = 8$

(ii) Squares of terms : $\sum T_n^2 = \dfrac{a^2}{1 - r^2}$; odd-place terms ka sum $= \dfrac{a}{1 - r^2}$

(iii) Har term = uske baad ke **saare** terms ka sum $\Rightarrow \boxed{r = \dfrac{1}{2}}$

☆☆ (iv) a, b, c in G.P. aur $a + b + c = xb \Rightarrow x = r + \dfrac{1}{r} + 1 \Rightarrow$

$\boxed{x \notin (-1, 3)}$

(v) p-th, q-th, r-th terms A, B, C of a G.P. : $\boxed{A^{q-r}\,B^{r-p}\,C^{p-q} = 1}$

A.P. aur G.P. dono mix

a, b, c in A.P. aur a^2, b^2, c^2 in G.P. $\Rightarrow a + c = 2b$, $b^2 = \pm ac$

$\rightarrow b^2 = +ac \Rightarrow a = b = c$; $b^2 = -ac \Rightarrow a^2 + 6ac + c^2 = 0 \Rightarrow \dfrac{c}{a} = -3 \pm 2\sqrt{2}$

Dono case alag likho, sirf $b^2 = ac$ lena adhoora hai.

Infinite G.P. ke saath limit

$\rightarrow x = a + a^2 + a^3 + \ldots$ ($|a| < 1$) $\Rightarrow x = \dfrac{a}{1 - a} \Rightarrow a = \dfrac{x}{1 + x}$

$\rightarrow \lim\limits_{n\to\infty}\left(\dfrac{1}{2} + \dfrac{1}{4} + \cdots + \dfrac{1}{2^n}\right) = 1$; $\sum\limits_{n=0}^{\infty} \dfrac{x^n}{3^n}$ converge $\iff |x| < 3$

S_∞ ka formula tabhi lagao jab $|r| < 1$ ho, warna series diverge karti hai.

Harmonic Progression (H.P.) : reciprocals are in A.P.

Agar $\frac{1}{T_1}, \frac{1}{T_2}, \ldots$ ka first term A aur diff D ho, toh :

☆☆ $T_n = \dfrac{1}{A + (n-1)D}$ H.P. ka sum formula nahi hota!

☆ Trick : H.P. ka koi bhi question aaye $\rightarrow$ seedha reciprocal lo, A.P. bana do.

a, b, c in H.P. $\iff \dfrac{2}{b} = \dfrac{1}{a} + \dfrac{1}{c} \iff b = \dfrac{2ac}{a+c}$

H.M. insert karo : n H.M.s between $a, b \Rightarrow$ reciprocals me n A.M.s :

$$D = \frac{\frac{1}{b} - \frac{1}{a}}{n+1} = \frac{a-b}{(n+1)ab}$$

Ex: 7th term of H.P. is $\frac{1}{10}$, 12th is $\frac{1}{25}$. Find 19th. $\Rightarrow$ A.P. : $T_7 = 10,\ T_{12} = 25$

$5d = 15 \Rightarrow d = 3 \ \Rightarrow \ T_{19} = 10 + 12(3) = 46 \ \Rightarrow$ H.P. ka 19th term $= \boxed{\frac{1}{46}}$

A.M., G.M., H.M. of two positive numbers a, b

$$A = \frac{a+b}{2} \quad ; \quad G = \sqrt{ab} \quad ; \quad H = \frac{2ab}{a+b}$$

☆☆ $A \geq G \geq H$ (equal iff $a = b$) ; $\boxed{G^2 = AH}$ $\Rightarrow A, G, H$ in G.P.

(i) Quadratic with roots a, b : $\boxed{x^2 - 2Ax + G^2 = 0}$ $\Rightarrow a, b = A \pm \sqrt{A^2 - G^2}$

(ii) $A : G = m : n \ \Rightarrow \ a : b = \left(m + \sqrt{m^2 - n^2}\right) : \left(m - \sqrt{m^2 - n^2}\right)$

(iii) n positive numbers : $\dfrac{a_1 + \cdots + a_n}{n} \geq (a_1 a_2 \cdots a_n)^{1/n} \geq \dfrac{n}{\frac{1}{a_1} + \cdots + \frac{1}{a_n}}$

Ex: A.M. $= 10$, G.M. $= 8 \ \Rightarrow$ numbers $= 10 \pm \sqrt{100 - 64} = \boxed{16,\ 4}$; $H = \dfrac{64}{10} = 6.4$

H.P. ke kuch aur kaam ke points

(i) a, b, c in H.P. $\iff \dfrac{a}{c} = \dfrac{a-b}{b-c}$; $\boxed{\dfrac{b+a}{b-a} + \dfrac{b+c}{b-c} = 2}$

(ii) a, b, c in H.P. $\Rightarrow \dfrac{b+c}{a}, \dfrac{c+a}{b}, \dfrac{a+b}{c}$ in A.P. $\left(\frac{S}{a} - 1 \text{ wali trick}\right)$

Use of A.M. $\geq$ G.M. (max / min wale sawal)

→ Sum fixed ho toh product **maximum** → Product fixed ho toh sum **minimum**

$x > 0 : x + \frac{1}{x} \geq 2$; $x + y = 10 \Rightarrow xy \leq 25$; $(a+b+c)\left(\frac{1}{a}+\frac{1}{b}+\frac{1}{c}\right) \geq 9$

$2^x + 2^{1-x} \geq 2\sqrt{2^x \cdot 2^{1-x}} = 2\sqrt{2}$ (min value)

n means between a and b (A_i, G_i, H_i)

$\boxed{\sum A_i = \frac{n(a+b)}{2}}$; $\boxed{\prod G_i = (ab)^{n/2}}$; $\boxed{\sum \frac{1}{H_i} = \frac{n(a+b)}{2ab}}$

Symmetry : $A_i + A_{n+1-i} = a+b$; $G_i G_{n+1-i} = ab$; ☆ $\boxed{A_i \cdot H_{n+1-i} = ab}$

Ex: $a=2, b=7, n=4 : A = 3,4,5,6$; $H = \frac{7}{3}, \frac{14}{5}, \frac{7}{2}, \frac{14}{3} \Rightarrow A_1 H_4 = 14 = ab$ ✓

Inequalities to remember (Advanced)

(i) **Weighted A.M.-G.M.** : $\frac{w_1 a_1 + \cdots + w_n a_n}{w_1 + \cdots + w_n} \geq \left(a_1^{w_1} \cdots a_n^{w_n}\right)^{\frac{1}{\sum w}}$

(ii) **Cauchy-Schwarz** : $\boxed{(\sum a_i b_i)^2 \leq (\sum a_i^2)(\sum b_i^2)}$

(iii) **Power mean** : $m > 1 \Rightarrow \left(\frac{a^m + b^m}{2}\right)^{\frac{1}{m}} \geq \frac{a+b}{2}$

(iv) $a, b, c > 0$: $a^3 + b^3 + c^3 \geq 3abc$; $(a+b)(b+c)(c+a) \geq 8abc$

(v) **Nesbitt** : $\frac{a}{b+c} + \frac{b}{c+a} + \frac{c}{a+b} \geq \frac{3}{2}$; $n! < \left(\frac{n+1}{2}\right)^n$

Weights ka jugaad $x^p y^q$ ka max

Ex: $x, y > 0$, $x + y = 5$. Max of $x^2 y^3$? $\Rightarrow$ x ko do tukde $\frac{x}{2}, \frac{x}{2}$ aur y ko teen $\frac{y}{3}, \frac{y}{3}, \frac{y}{3}$ karo :

$$\frac{\frac{x}{2} + \frac{x}{2} + \frac{y}{3} + \frac{y}{3} + \frac{y}{3}}{5} = 1 \geq \left(\frac{x^2 y^3}{4 \cdot 27}\right)^{1/5} \Rightarrow x^2 y^3 \leq \boxed{108} \quad (\text{at } x=2, y=3)$$

Ex: $x > 0$. Min of $3x + \frac{4}{x^2}$? $\Rightarrow$ $\frac{3x}{2} + \frac{3x}{2} + \frac{4}{x^2} \geq 3\sqrt[3]{\frac{9x^2}{4} \cdot \frac{4}{x^2}} = \boxed{3\sqrt[3]{9}}$

Trick : x ko aise todo ki product me x cancel ho jaye (powers $x \cdot x \cdot x^{-2}$).

Ex: 2 aur 12 ke beech 3 H.M.s : reciprocals $\frac{1}{2}, \ldots, \frac{1}{12}$ me $D = \frac{\frac{1}{12} - \frac{1}{2}}{4} = -\frac{5}{48}$

A.P. : $\frac{19}{48}, \frac{14}{48}, \frac{9}{48}$ $\Rightarrow$ H.M.s $\boxed{\frac{48}{19}, \frac{24}{7}, \frac{16}{3}}$

Summation by Sigma Operator

→ $\sum(a_r \pm b_r) = \sum a_r \pm \sum b_r \quad ; \quad \sum c\,a_r = c\sum a_r$

→ $\sum_{r=1}^{n} c = nc$ c nahi, nc! (constant n baar jodo)

→ $\sum_{r=m}^{n} f(r) = \sum_{r=1}^{n} f(r) - \sum_{r=1}^{m-1} f(r) \quad ; \quad \text{terms} = n - m + 1$

Standard results

☆ $\sum r = \dfrac{n(n+1)}{2}$

☆ $\sum r^2 = \dfrac{n(n+1)(2n+1)}{6}$

☆ $\sum r^3 = \left[\dfrac{n(n+1)}{2}\right]^2$

$\sum(2r-1) = n^2$

$\sum(2r-1)^2 = \dfrac{n(2n-1)(2n+1)}{3}$

$\sum r(r+1) = \dfrac{n(n+1)(n+2)}{3}$

Method : pehle T_r (general term) likho, expand karo, phir $\sum$ lagao.

Ex: $1\cdot 3 + 2\cdot 5 + 3\cdot 7 + \ldots$ (n terms) $\Rightarrow T_r = r(2r+1)$

$S_n = 2\sum r^2 + \sum r = \dfrac{n(n+1)(2n+1)}{3} + \dfrac{n(n+1)}{2} = \dfrac{n(n+1)(4n+5)}{6}$

Aur kuch Sigma results

☆ $\sum r^4 = \dfrac{n(n+1)(2n+1)(3n^2+3n-1)}{30} \quad ; \quad \displaystyle\sum_{r=1}^{n} \frac{1}{1+2+\cdots+r} = \frac{2n}{n+1}$

Double sums : $\displaystyle\sum_{i=1}^{n}\sum_{j=1}^{i} j = \frac{n(n+1)(n+2)}{6} \quad ; \quad \sum_{i,j=1}^{n}(i+j) = n^2(n+1)$

$\displaystyle\sum_{i,j=1}^{n} \min(i,j) = \frac{n(n+1)(2n+1)}{6} \quad ; \quad \sum_{1\le i<j\le n}(i+j) = \frac{(n-1)n(n+1)}{2}$

Ex: $3 + 7 + 13 + 21 + \ldots$ (n terms) : $T_r = r^2 + r + 1 \Rightarrow S_n =$

$\dfrac{n(n+1)(2n+1)}{6} + \dfrac{n(n+1)}{2} + n = \dfrac{n(n^2+3n+5)}{3}$

Ex: $1\cdot 2^2 + 2\cdot 3^2 + 3\cdot 4^2 + \ldots$: $T_r = r(r+1)^2 = r^3 + 2r^2 + r$

$S_n = \dfrac{n^2(n+1)^2}{4} + \dfrac{n(n+1)(2n+1)}{3} + \dfrac{n(n+1)}{2} =$

$\dfrac{n(n+1)(n+2)(3n+5)}{12}$

Arithmetico-Geometric Progression (A.G.P.)

$a,\ (a+d)r,\ (a+2d)r^2, \ldots$ $\quad \boxed{T_n = [a+(n-1)d]\,r^{n-1}}$

☆☆ **Method :** S likho, $\times r$ karo, shift karke subtract $\Rightarrow S(1-r) = \ldots$

$$\boxed{S_n = \frac{a}{1-r} + \frac{dr\,(1-r^{n-1})}{(1-r)^2} - \frac{[a+(n-1)d]\,r^n}{1-r}}$$

$$\boxed{\text{☆}\ \ S_\infty = \frac{a}{1-r} + \frac{dr}{(1-r)^2} \quad (|r|<1)}$$

Ex: $1 + \frac{2}{2} + \frac{3}{4} + \frac{4}{8} + \ldots$; $\quad a=1,\ d=1,\ r=\frac{1}{2} \ \Rightarrow\ S_\infty = 2 + \frac{1/2}{1/4} = \boxed{4}$

Also yaad rakho : $1 + 2x + 3x^2 + \cdots = \dfrac{1}{(1-x)^2}$

Recurrence relations (Advanced)

Type-1 : $T_{n+1} = pT_n + q$ (fixed point $\alpha = \frac{q}{1-p}$)

$$\boxed{T_n - \alpha = p^{n-1}(T_1 - \alpha)}$$

Ex: $T_1 = 1,\ T_{n+1} = 2T_n + 3 \Rightarrow \alpha = -3 \ \Rightarrow\ T_n = 2^{n+1} - 3$

Type-2 : $T_{n+2} = pT_{n+1} + qT_n$ (Fibonacci type)

$T_n = \lambda^n$ try karo : $\boxed{\lambda^2 = p\lambda + q} \ \Rightarrow\ T_n = A\lambda_1^n + B\lambda_2^n$

Ex: $T_{n+2} = 5T_{n+1} - 6T_n,\ T_1 = 1, T_2 = 5 \ \Rightarrow\ \lambda = 2, 3 \ \Rightarrow\ T_n = \boxed{3^n - 2^n}$

Type-3 : $T_{n+1} - T_n = f(n) \ \Rightarrow\ T_n = T_1 + \sum_{k=1}^{n-1} f(k)$

Ex: $a_{n+1} - a_n = n,\ a_1 = 1 \ \Rightarrow\ a_n = 1 + \dfrac{n(n-1)}{2}$

Type-4 : periodic sequence $a_{n+1} = \dfrac{1}{1-a_n} \ \Rightarrow$ **period 3**

$a_1 = 2$: $2, -1, \frac{1}{2}, 2, -1, \frac{1}{2}, \ldots \ \Rightarrow\ a_{100} = a_1 = 2$ (100 mod 3 = 1)

☆☆ **Type-5 :** telescoping recursion $a_{n+1} = a_n^2 - a_n + 1$

$$a_{n+1} - 1 = a_n(a_n - 1) \ \Rightarrow\ \frac{1}{a_n} = \frac{1}{a_n - 1} - \frac{1}{a_{n+1} - 1}$$

$$\boxed{\sum_{k=1}^{n} \frac{1}{a_k} = \frac{1}{a_1 - 1} - \frac{1}{a_{n+1} - 1}} \quad a_1 = 2 \text{ ho toh } S_\infty = 1$$

Sum of Series by Method of Difference

Agar $T_r = V_r - V_{r-1}$ likh sako (telescoping), toh $\boxed{S_n = V_n - V_0}$ beech ke terms cancel!

Type-1 : factors A.P. me (fraction)

$$\frac{1}{r(r+d)} = \frac{1}{d}\left(\frac{1}{r} - \frac{1}{r+d}\right) \quad \text{Ex: } \sum_{r=1}^{n} \frac{1}{r(r+1)} = 1 - \frac{1}{n+1} = \frac{n}{n+1}$$

$$\frac{1}{r(r+1)(r+2)} = \frac{1}{2}\left[\frac{1}{r(r+1)} - \frac{1}{(r+1)(r+2)}\right] \Rightarrow$$

$$\boxed{S_n = \frac{1}{4} - \frac{1}{2(n+1)(n+2)}}$$

Type-2 : consecutive numbers ka product (polynomial)

$$r(r+1)(r+2) = \frac{1}{4}\left[r(r+1)(r+2)(r+3) - (r-1)r(r+1)(r+2)\right]$$

$$\Rightarrow \boxed{\sum r(r+1)(r+2) = \frac{n(n+1)(n+2)(n+3)}{4}}$$ (ek aur factor, ek se divide)

Type-3 : surds, factorials, inverse tan

$\rightarrow \dfrac{1}{\sqrt{r+1}+\sqrt{r}} = \sqrt{r+1} - \sqrt{r} \Rightarrow S_n = \sqrt{n+1} - 1$

$\rightarrow r \cdot r! = (r+1)! - r! \Rightarrow S_n = (n+1)! - 1 \quad ; \quad \dfrac{r}{(r+1)!} = \dfrac{1}{r!} - \dfrac{1}{(r+1)!}$

☆☆ $\rightarrow \tan^{-1}\dfrac{1}{1+r(r+1)} = \tan^{-1}(r+1) - \tan^{-1} r \Rightarrow S_n = \tan^{-1}(n+1) - \dfrac{\pi}{4}$

Type-4 : terms ke differences A.P. / G.P. banate hain

$\rightarrow S = T_1 + T_2 + \cdots + T_n$ aur $S = \quad T_1 + \cdots + T_{n-1} + T_n$ (ek jagah shift)

$\Rightarrow$ subtract

$\rightarrow$ ya seedha $T_n = T_1 + (\text{sum of first } n-1 \text{ differences})$

Ex: $2 + 5 + 10 + 17 + \ldots$: differences $3, 5, 7, \ldots$ (A.P.) $\Rightarrow T_n = 2 + \dfrac{n-1}{2}[6 + 2(n-2)] = n^2 + 1$

Ex: $1 + 3 + 7 + 15 + 31 + \ldots$: differences $2, 4, 8, 16$ (G.P.) $\Rightarrow T_n = 2^n - 1$

$S_n = \sum(2^r - 1) = 2^{n+1} - 2 - n$

Sum of Some Special Series

☆ (i) Alternate signs : $\boxed{1^2 - 2^2 + 3^2 - \cdots + (-1)^{n-1} n^2 = (-1)^{n-1} \frac{n(n+1)}{2}}$

Trick : do-do terms ka pair banao, $(1^2 - 2^2) = -3,\ (3^2 - 4^2) = -7, \ldots$

(ii) Repeated digits : $9 + 99 + 999 + \ldots$ (n terms) $= \frac{10(10^n - 1)}{9} - n$

(iii) Cubes of odd numbers : $1^3 + 3^3 + \cdots + (2n-1)^3 = n^2(2n^2 - 1)$

(iv) Pairs : $\boxed{\sum_{i<j} a_i a_j = \frac{(\sum a_i)^2 - \sum a_i^2}{2}}$

Ex: $\sum_{1 \le i < j \le 3} ij = 2 + 3 + 6 = 11 = \frac{36 - 14}{2}$ ✓

JEE Special Types

Type-1 : Common terms of two A.P.s

Common terms khud A.P. banate hain, jiska $\boxed{\text{common diff} = \text{LCM}(d_1, d_2)}$

Ex: $3, 7, 11, \ldots, 407$ aur $2, 9, 16, \ldots, 709$ ke common terms ?

$x \equiv 3 \pmod 4,\ \ x \equiv 2 \pmod 7 \ \Rightarrow\ x = 23, 51, \ldots$ (diff 28)

$23 + 28k \le 407 \Rightarrow k \le 13.7 \ \Rightarrow\ k = 0, \ldots, 13 \ \Rightarrow$ (14 terms)

Type-2 : Roots of a cubic in A.P. / G.P.

$x^3 + px^2 + qx + s = 0$ Roots in A.P. : $\alpha - d,\ \alpha,\ \alpha + d \ \Rightarrow\ \boxed{\alpha = -\frac{p}{3}}$

Roots in G.P. : $\frac{\alpha}{r},\ \alpha,\ \alpha r \ \Rightarrow\ \boxed{\alpha^3 = -s}$ (middle root pehle nikalo)

Roots in H.P. : $y = \frac{1}{x}$ lo $\Rightarrow$ new equation ke roots A.P. me

Type-3 : Agar a, b, c A.P. me hain

$\rightarrow\ b + c - a,\ \ c + a - b,\ \ a + b - c$ bhi A.P. me (kyunki $= S - 2a,\ S - 2b,\ S - 2c$)

$\rightarrow\ \frac{1}{bc}, \frac{1}{ca}, \frac{1}{ab}$ bhi A.P. me (kyunki a, b, c ko abc se divide kiya)

$\rightarrow\ a, b, c$ A.P. me aur a, b, c ka sum diya ho $\Rightarrow$ b seedha mil jata hai ($3b = \text{sum}$)

Type-4 : Numbers 1 to N jo p ya q se divisible nahi

Inclusion-exclusion : $\Sigma_{\text{all}} - \Sigma_p - \Sigma_q + \Sigma_{pq}$ (har ek A.P. hai)

Ex: 1 to 100 me 3 ya 5 se na bate : $5050 - 1683 - 1050 + 315 =$ (2632)

Good Ques. JEE Main style

Q1. $\sum_{r=1}^{\infty} \frac{1}{r(r+2)} =$ ------ Ans : 3/4

Q2. A.M. of two positive numbers is 5, H.M. is $\frac{16}{5}$. The numbers are ------ Ans : 8, 2

Q3. $1 + \frac{4}{5} + \frac{7}{25} + \frac{10}{125} + \ldots \infty =$ ------ Ans : 35/16

Q4. In an A.P., $T_3 = 7$ and $T_7 = 3T_3 + 2$. Find S_{20}. Ans : 740

Q5. Sum of all three-digit numbers divisible by 7 is ------ Ans : 70336

Q6. Sum to infinity of a G.P. is 15 and sum of squares of its terms is 45. Find a, r. Ans : 5, $\frac{2}{3}$

Q7. $\sum_{r=1}^{10} r(r+2) =$ ------ Ans : 495

Q8. Three positive numbers in G.P. have product 216 and sum of products in pairs 156. The numbers are ------ Ans : 2, 6, 18

Good Ques. (Advanced) JEE Adv style

Q1. $a < b < c$ in A.P., a^2, b^2, c^2 in G.P., $a + b + c = \frac{3}{2}$. Find a. Ans : $\frac{1}{2} - \frac{1}{\sqrt{2}}$

Q2. $a_1 = 2,\ a_{n+1} = a_n^2 - a_n + 1$. Find $\sum_{k=1}^{\infty} \frac{1}{a_k}$. Ans : 1

Q3. $x, y > 0,\ x + y = 5$. Maximum value of x^2y^3 is ------ Ans : 108

Q4. S_n of an A.P. is such that $\frac{S_{2n}}{S_n}$ is independent of n. Then $\frac{S_{3n}}{S_n} =$ ------ Ans : 9

Q5. $\sum_{n=1}^{\infty} \frac{n^2}{2^n} =$ ------ Ans : 6

Q6. $\lim_{n\to\infty} \sum_{r=1}^{n} \frac{r}{r^4 + r^2 + 1} =$ ------ Ans : $\frac{1}{2}$

Kaunsa method lagaun ?

T_n me constant difference	$\Rightarrow$ A.P. formula
constant ratio / $(a+(n-1)d)r^{n-1}$	$\Rightarrow$ G.P. / A.G.P. $(S-rS)$
reciprocals A.P. me	$\Rightarrow$ H.P. : flip karo
T_r polynomial in r	$\Rightarrow \sum r,\ \sum r^2,\ \sum r^3$
$\frac{1}{r(r+d)}$, $r\cdot r!$, $\sqrt{r+1}-\sqrt{r}$	$\Rightarrow$ Method of Difference
max / min, $a+b$ ya ab fixed	$\Rightarrow$ A.M. $\geq$ G.M. $\geq$ H.M.
T_{n+1} in terms of T_n	$\Rightarrow$ Recurrence : type pehchano

Last minute formula strip

A.P. $T_n = a+(n-1)d$

$S_n = \frac{n}{2}[2a+(n-1)d]$; $d = \frac{b-a}{n+1}$

G.P. $T_n = ar^{n-1}$; $S_\infty = \frac{a}{1-r}$

$S_n = a\frac{r^n-1}{r-1}$; $r = \left(\frac{b}{a}\right)^{\frac{1}{n+1}}$

H.P. reciprocals in A.P.

$H = \frac{2ab}{a+b}$; $G^2 = AH$

$A \geq G \geq H$

Sigma $\sum r = \frac{n(n+1)}{2}$

$\sum r^2 = \frac{n(n+1)(2n+1)}{6}$

$\sum r^3 = \left[\frac{n(n+1)}{2}\right]^2$

A.G.P. $S_\infty = \frac{a}{1-r} + \frac{dr}{(1-r)^2}$

$T_n = T_1 + \sum(\text{differences})$

Recurrence $T_{n+1} = pT_n + q$:

$T_n - \alpha = p^{n-1}(T_1 - \alpha)$, $\alpha = \frac{q}{1-p}$

Dhyan rakho (common galtiyan)

☆ $T_n = S_n - S_{n-1}$ me T_1 alag se check karo. S_∞ sirf $|r|<1$ pe.

☆ $A \geq G \geq H$ sirf **positive** numbers ke liye. $\sum c = nc$, sirf c nahi.

☆ Terms count : $n-m+1$. S_n = given hone par n ke **do** answer bhi aa sakte hain.

Hints (separately) JEE Main style

Q1 1/(r(r+2)) ko partial fractions me todo ; terms do-do ke gap se cancel honge.

Q2 Pehle $G^2 = AH$ se G nikalo, phir roots ke liye $x^2 - 2Ax + G^2 = 0$ ya $A \pm \sqrt{A^2 - G^2}$.

Q3 Pehchano : A.G.P. hai ($a = 1$, $d = 3$, $r = \frac{1}{5}$). $S_\infty = \frac{a}{1-r} + \frac{dr}{(1-r)^2}$.

Q4 T_7 pehle nikalo, $4d = T_7 - T_3$ se d, phir a, last me $S_{20} = 10(2a + 19d)$.

Q5 A.P. $105, 112, \ldots, 994$: pehle terms ginti karo, phir $\frac{n}{2}(\text{first} + \text{last})$.

Q6 Do equations likho ; $(S_\infty)^2 \div \sum T^2 = \frac{1+r}{1-r}$ se seedha r mil jata hai.

Q7 $r(r+2) = r^2 + 2r$ expand karo, phir $\sum r^2$ aur $\sum r$ ke standard formulas.

Q8 $\frac{a}{r}, a, ar$ lo : product se a, pairs ke sum se $r + \frac{1}{r}$.

Hints (Advanced) JEE Adv style

Q1 Sum se $b = \frac{1}{2}$. $b^2 = ac$ aur $b^2 = -ac$ dono case dekho ; $a < b < c$ hone se $d \neq 0$.

Q2 $a_{n+1} - 1 = a_n(a_n - 1)$ likho, $\frac{1}{a_n}$ ko do fractions ka difference banao ; $a_n \to \infty$.

Q3 x ko $\frac{x}{2}, \frac{x}{2}$ aur y ko $\frac{y}{3}, \frac{y}{3}, \frac{y}{3}$ todo ; paanchon ka sum fixed hai, A.M.-G.M. lagao.

Q4 $\frac{S_{2n}}{S_n}$ ko a, d, n me likho ; n se independent tabhi jab $a = \frac{d}{2}$.

Q5 $n^2 = n(n-1) + n$ likho ya $\sum x^n$ ko do baar differentiate karo, phir $x = \frac{1}{2}$.

Q6 $r^4 + r^2 + 1 = (r^2 - r + 1)(r^2 + r + 1)$; dono factors ka difference $2r$ hai.

Advanced me kya dhyan rakhein

☆ "Sabhi n ke liye" condition aaye $\to$ coefficients compare karo ($a = \frac{d}{2}$ jaisa).

☆ Infinite sum se pehle **convergence** check karo ($|r| < 1$).

☆ Telescoping dikhe nahi raha ? partial fractions ya recursion se $\frac{1}{a_n}$ todo.

☆ A.M.-G.M. lagane ke baad **equality possible hai ya nahi** zaroor check karo.